

Climate Change Overview

What is Climate Change?

Climate change refers to long-term shifts in global temperatures and weather patterns. While some changes are natural, since the 1800s human activities have been the main driver, primarily through burning fossil fuels like coal, oil, and natural gas.

Greenhouse Effect

The greenhouse effect occurs when gases like carbon dioxide and methane trap heat from the sun in the atmosphere. This natural process is essential for life, but increased greenhouse gas emissions are intensifying it, causing global temperatures to rise.

Current Impact

Global temperatures have risen approximately 1.1 degrees Celsius above pre-industrial levels. This has led to more frequent extreme weather events, rising sea levels, melting polar ice caps, and disruption of ecosystems worldwide.

Solutions

Addressing climate change requires transitioning to renewable energy sources like solar and wind power, improving energy efficiency, reducing deforestation, and adopting sustainable agricultural practices. International cooperation through agreements like the Paris Accord is critical.